**Department- Information Technology**
**SY-IT A**
**Sem-III**
**Name: Rashi Dawda**
**Roll no -23**

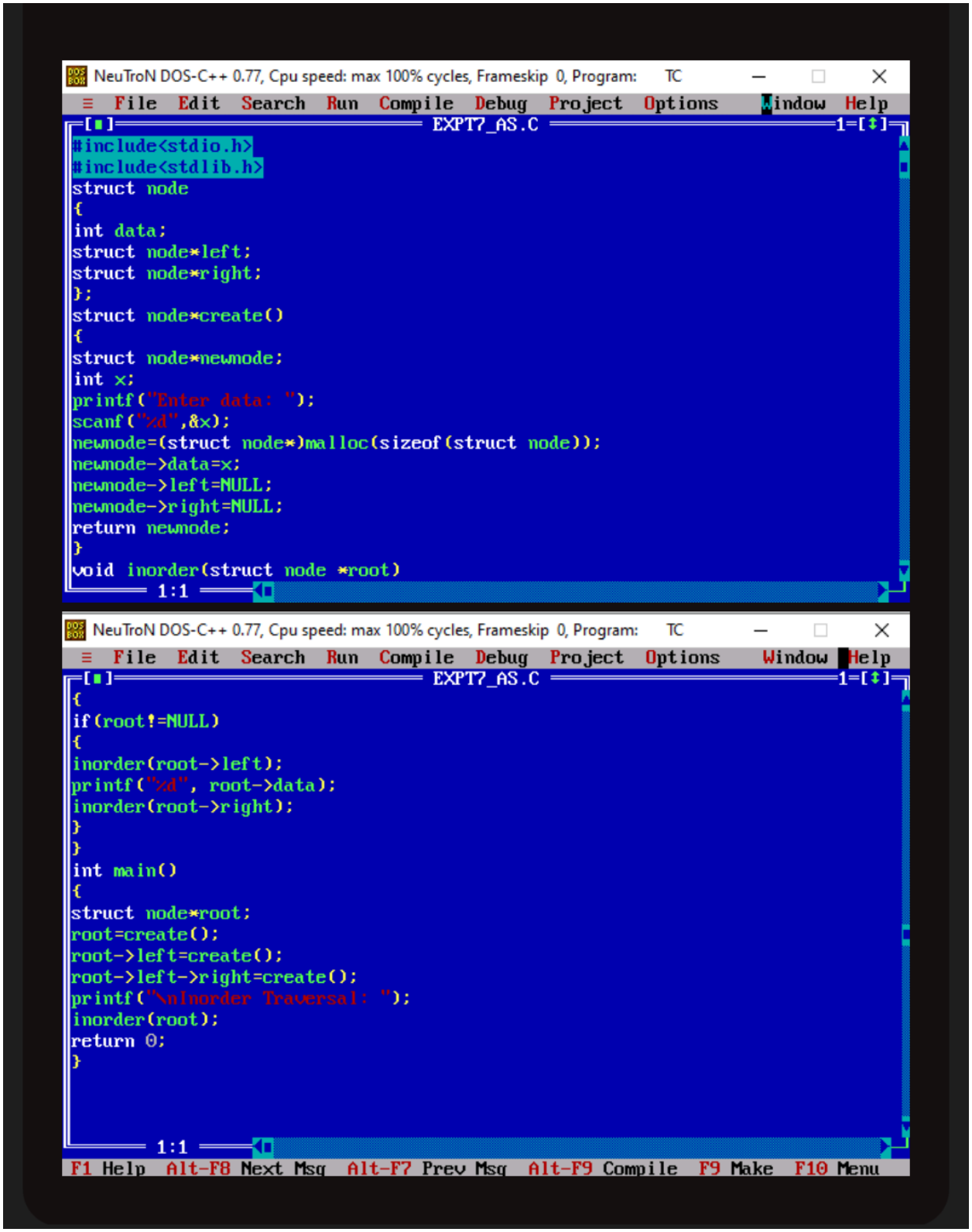
## Experiment no-6

**Aim:**
To implement a Binary Tree and perform its Inorder, Preorder, and Postorder traversals.

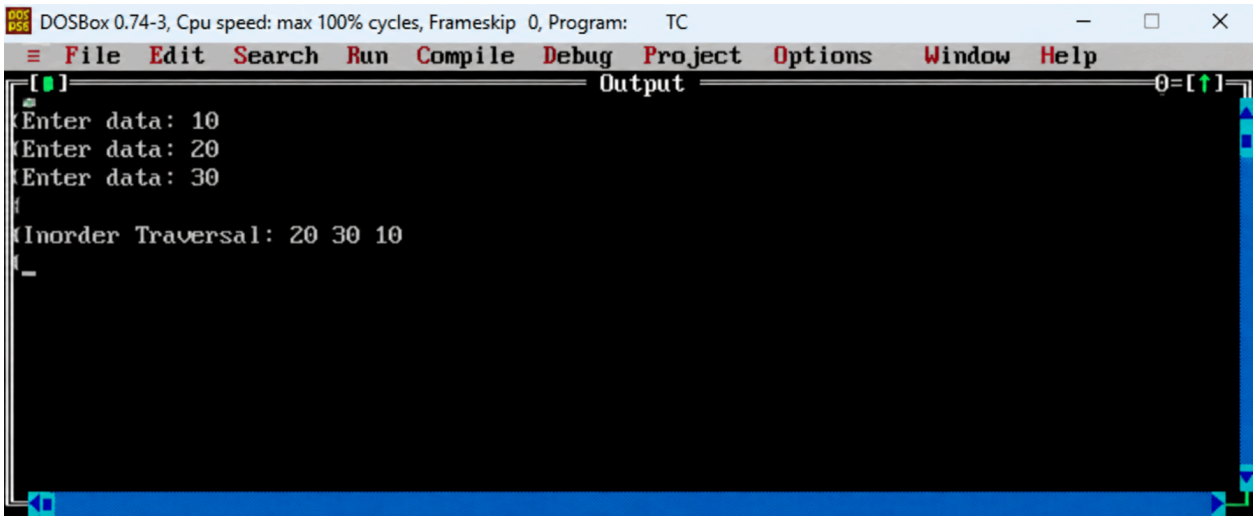
**Objectives:**
To understand the concept of Binary Trees.
To create and insert nodes in a Binary Tree.
To perform different tree traversals.
To understand its real-world applications.

**Theory:**
A Binary Tree is a hierarchical data structure in which each node has at most two children: left and right.
Inorder: Left → Root → Right
Preorder: Root → Left → Right
Postorder: Left → Right → Root
Binary Trees are used in applications such as file systems, decision trees, and expression evaluation.

**Code**:


NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXPT7_AS.C
#include<stdio.h>
#include<stdlib.h>
struct node
{
int data;
struct node*left;
struct node*right;
};
struct node*create()
{
struct node*newmode;
int x;
printf("Enter data: ");
scanf("%d",&x);
newmode=(struct node*)malloc(sizeof(struct node));
newmode->data=x;
newmode->left=NULL;
newmode->right=NULL;
return newmode;
}
void inorder(struct node *root)
1:1
NeuTroN DOS-C++ 0.77, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
EXPT7_AS.C
{
if(root!=NULL)
{
inorder(root->left);
printf("%d", root->data);
inorder(root->right);
}
}
int main()
{
struct node*root;
root=create();
root->left=create();
root->left->right=create();
printf("\nInorder Traversal: ");
inorder(root);
return 0;
}
1:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

**Output**:




**Conclusion:**
Thus, a Binary Tree was successfully implemented and its different traversals were performed.

**Outcome:**
The experiment helped us understand Binary Tree operations and traversal techniques and their practical applications.